Registration No	B	I	U	C	2	0	2	5	M	S	C	C	A	I	T	6	6	8	0	
Student Name	D	A	L	A	L		Z	A	M	Z	A	M								
Program Name	M	S	C	A	I															
Course Code	M	S	C	A	I	C	B	R	M											
Date	2	5	1	2	2	0	2	5												
Type of Assessment	☐	**Internal Exam**						■	**Final Project**											
Examiner Name								**Examiner Signature with Date**												

AI grading system for Arabic handwritten exams
By: Dalal Adnan

Abstract

The manual grading of handwritten examinations is a time-consuming and subjective process, particularly for Arabic scripts, which are cursive, context-dependent, and highly variable in handwriting style. This study proposes an AI-based grading system for Arabic handwritten exam papers that integrates optical character recognition (OCR) with semantic answer evaluation to support automated and semi-automated grading. The proposed system processes scanned or photographed handwritten responses, converts them into machine-readable text, retrieves model answers, and evaluates student responses using similarity-based scoring techniques. The primary objective is to assess whether AI-assisted grading can achieve a level of accuracy comparable to human graders while significantly reducing grading time. The system is evaluated using metrics such as grading agreement with human assessors, score deviation, recognition accuracy, and time efficiency. Additionally, the study analyzes the impact of handwriting quality and image characteristics on grading performance. The findings aim to demonstrate the feasibility of deploying AI-assisted grading as a decision-support tool in educational environments, particularly for Arabic language assessments.

Introduction

Handwritten examinations remain widely used in Arabic-language education, particularly in language and theory-based subjects. While handwritten responses allow expressive answers, manual grading is time-consuming, inconsistent, and prone to subjectivity, especially in large-scale educational environments. These challenges are amplified in Arabic due to the cursive nature of the script, context-dependent letter shapes, diacritics, and high variability in handwriting styles. Recent advances in artificial intelligence (AI), including deep learning–based optical character recognition (OCR) and natural language processing (NLP), have enabled more effective processing of handwritten text and semantic analysis, creating opportunities for AI-assisted grading systems.

Despite these advances, most automated grading solutions focus on multiple-choice or typed responses, leaving Arabic handwritten exam grading largely manual. Existing Arabic handwriting recognition systems are sensitive to handwriting quality and image

acquisition conditions, which directly affect grading reliability. There is currently a lack of integrated frameworks that combine Arabic handwriting recognition with semantic answer evaluation to support practical AI-assisted grading in real educational contexts. This study investigates whether an AI-assisted grading system can achieve grading accuracy comparable to human graders while reducing grading time for Arabic handwritten exam answers. The research hypothesizes that AI-assisted grading can significantly improve efficiency while maintaining strong agreement with human scores, and that grading performance is strongly influenced by handwriting recognition accuracy, image quality, and handwriting style.
The objectives of this work are to design an AI-assisted grading framework for Arabic handwritten responses, evaluate its grading accuracy against human assessors, measure efficiency gains, and analyze the key factors affecting system performance. By addressing these objectives, this research contributes to the advancement of educational AI and provides practical insights into deploying scalable and consistent grading support tools for Arabic-language assessment.

Literature Review

A. End-to-End Automated Grading Pipelines (Handwriting → Text → Score)
Recent work is moving from “single components” toward **complete automated grading pipelines**. Kishor *et al.* propose an automatic answer-sheet evaluation stack that integrates OCR with NLP for grading, positioning the solution as a practical institutional tool rather than a purely academic model. ScienceDirect+1 This direction is aligned with real school constraints, but such systems remain strongly dependent on OCR reliability and typically assume relatively controlled inputs. A comparable “education workflow” mindset appears in Jung *et al.*, who discuss operationalizing automated scoring (including rubric framing and evaluation), reinforcing the idea that scoring systems must be designed with deployment realities—not just model accuracy—in mind. ScienceDirect
Gap: These pipeline papers rarely target **Arabic handwritten exam pages end-to-end with rubric-based partial credit**, and they seldom quantify school-level operational impact (teacher time saved, dispute rate, standardization across graders) in a rigorous experimental design.

B. Arabic Handwriting Recognition: Segmentation, Ligatures, and Layout
Arabic handwriting recognition (AHR) remains difficult due to **connected script, ligatures, and large intra-writer variability**. Berriche *et al.* address a key bottleneck—segmentation—using a CNN-based approach combined with a graph-theory algorithm

to split Arabic handwritten words into sub-words, improving the reliability of downstream recognition. ScienceDirect+1 Al Hamad and Shehab target segmentation errors caused by Arabic ligatures via a ligature-detection technique, showing that explicitly modeling Arabic script structure improves segmentation quality. ScienceDirect On the recognition side, Alghyaline evaluates optimized CNN architectures for Arabic handwritten character recognition, indicating continued value in careful model/feature design even in the era of transformers. ScienceDirect

Beyond segmentation at the word level, **page understanding** is essential for exam papers. Aljiffry *et al.* apply Mask R-CNN for layout analysis on Arabic historical documents, demonstrating how deep layout segmentation can separate regions and support structured extraction. ScienceDirect+1 While the domain is "historical documents," the method is directly transferable to exam sheets where question blocks and answer areas must be isolated accurately.

Gap: Most AHR studies optimize **segmentation/recognition metrics** on curated datasets, but do not validate robustness on **real school scans** (skew, low contrast, photocopies, stamps, margins, mixed handwriting quality) nor measure how recognition errors propagate into **grading fairness** and **rubric accuracy**.

C. Transformer-Based Handwritten Text Recognition and Data Efficiency

Transformers are increasingly used to overcome limitations of CNN-RNN pipelines. Dhiaf *et al.* propose MSdocTr-Lite, a transformer designed for **full-page multi-script HTR**, leveraging curriculum learning to learn reading order at page level and reduce line segmentation dependence. ScienceDirect+1 This is highly relevant to exams where line segmentation is brittle. Li *et al.* introduce HTR-VT (Vision Transformer for HTR) and emphasize **data-efficient HTR** without requiring pretraining on massive extra datasets, which matters in education settings with limited labeled data. ScienceDirect

Gap: These advances are general HTR improvements, but they are not typically integrated with **assessment logic** (rubrics, concept coverage, partial marks). The missing layer is a validated bridge from **HTR confidence + uncertainty** into **grading decisions** and teacher review workflows.

D. Dataset Scarcity and Domain Mismatch (Why School Data Matters)

A consistent theme is that Arabic handwriting systems are constrained by **limited, domain-diverse labeled datasets**. Najam and Faizullah publish a scarce dataset for ancient Arabic handwritten text recognition, explicitly addressing the data bottleneck and supporting training/benchmarking. ScienceDirect+1 While valuable, ancient manuscripts differ substantially from student exam scripts (layout, writing purpose, vocabulary, noise patterns), creating a **domain mismatch** when transferring models into education.

Gap: There is still no widely adopted benchmark that combines **Arabic exam-page scans + transcriptions + question structure + rubric labels + final scores**. This is precisely where a school-based PoC can contribute a high-impact dataset and evaluation protocol.

E. Automated Scoring for Arabic Short Answers (Content Scoring Foundations)
Arabic automated scoring has a longer research history for short answers. Gomaa and Fahmy present an early Arabic short-answer scoring approach using multiple similarity measures (alone and in combination), establishing core principles such as matching student responses to reference answers while tolerating variation. ScienceDirect+1 More recent work, AraScore by ElNaka *et al.*, investigates response-based Arabic short answer scoring and evaluates approaches across datasets, indicating progress toward scalable scoring beyond simple lexical matching. ScienceDirect+2ScienceDirect+2 Nael *et al.* extend this direction into a deep learning-based system for Arabic short answer scoring, illustrating the shift toward modern Arabic NLP pipelines for assessment tasks. ScienceDirect
Gap: These scoring systems usually assume **typed text inputs**, meaning they do not address the dominant operational reality in many schools: **handwritten scripts**. The under-explored research space is the **joint optimization** of HTR + scoring under rubric constraints, including how to handle uncertainty (illegible writing, low confidence tokens) fairly.

F. Automated Essay Scoring (AES) for Arabic: Coherence, Hybrid Models, and LLMs
For Arabic essays, research is expanding from holistic scoring to quality dimensions. Machhout *et al.* propose a multi-task model to evaluate **coherence and cohesion** in Arabic essays, reflecting a move toward finer-grained writing quality assessment. ScienceDirect+1 Ezz *et al.* propose efficient Arabic essay scoring using hybrid models (feature selection/data optimization with learning models), aiming for performance with manageable complexity—useful for resource-constrained deployments. ScienceDirect LLM-based AES is also being tested. Ghazawi and Simpson benchmark multiple LLMs for Arabic essay scoring under zero-shot/few-shot/fine-tuning settings and show that prompting strategy and model choice matter, while smaller BERT-based systems can outperform LLMs on scoring agreement in some cases. ScienceDirect+2arXiv+2 Aydın extends the discussion by empirically evaluating automated scoring with rubric-based prompting in a low-resource language context, reinforcing that “AI scoring” must be judged by validity and reliability, not only by convenience. ScienceDirect+1

Gap: Arabic AES work generally does not incorporate **handwriting recognition** and usually lacks an end-to-end evaluation on **scanned handwritten exam essays**. Additionally, rubric alignment and explainability remain weak: schools require "why this score?" defensibility for student appeals.

G. Acceptance, Governance, and Education-Sector Deployment Constraints
Adoption is not purely technical; it is socio-technical. Duangmanee and Rungraungsilp compare teacher-led vs OCR-based handwriting assessment and analyze time efficiency, scoring accuracy, and teacher perspectives, emphasizing that teachers value control and trust. ScienceDirect+1 Pasco *et al.* similarly highlight rapid digital handwriting assessment and the practical evaluation of handwriting performance, reinforcing the operational push toward quicker feedback loops. ScienceDirect At the curriculum level, Sapawi *et al.* systematically review technology integration in Arabic language curriculum (2023–2025), showing growing demand but also fragmentation in evidence and implementation strategy. ScienceDirect+1 Alkaabi examines generative AI implementation and assessment in Arabic language teaching and reports instructor challenges around dialect variation, evaluation of AI-generated content, and academic integrity—issues that directly map to assessment governance. ScienceDirect+1 Jung *et al.* further support the necessity of responsible implementation practices when deploying automated scoring in real educational environments. ScienceDirect
Gap: Existing studies confirm the need for **human-in-the-loop** governance, but few provide a complete blueprint for Arabic handwritten exam grading: audit trails, teacher override, confidence thresholds, and appeal workflows tied directly to rubric evidence.

Research Gap and Value of the Proposed Work
Across these 20 studies, the field has strong progress in (1) Arabic handwriting recognition components, (2) Arabic short answer and essay scoring on typed inputs, and (3) early deployment/acceptance insights. However, a clear gap remains:

1. **No strong end-to-end validated system** for **Arabic handwritten exam pages → structured answers → rubric-based scoring** with teacher-approval workflows.
2. **No standardized dataset** combining **exam scans + transcription + rubric labels + scores** suitable for reproducible evaluation in the education sector.
3. Limited evidence on **fairness under uncertainty**: how low-confidence HTR and ambiguous handwriting affect rubric scoring and student outcomes.
4. Limited operational metrics: very few works quantify **time saved, inter-rater reliability improvements, and appeal/dispute rates** under realistic school constraints.

Methodology:

A. Research Design

This study adopts a quantitative, exploratory research design to investigate the feasibility and acceptance of an AI-based system for grading Arabic handwritten exam papers in the education sector. The research focuses on understanding stakeholders' perceptions, readiness, and key factors influencing the adoption of AI-assisted grading tools. A survey-based approach was selected due to its suitability for capturing attitudes and opinions from diverse educational roles.

B. Participants and Sampling [Appendix 3]

The target population consisted of stakeholders within the education sector, including teachers, academic coordinators, school leadership, IT personnel, and policy-level staff. A total of 35 respondents were included in the study. Participants represented multiple educational levels (primary, preparatory, secondary, and all levels combined) to ensure broad contextual coverage.

Due to the exploratory nature of the study and its focus on Proof of Concept (PoC) validation, convenience sampling was employed. This sampling strategy is commonly used in early-stage technology feasibility studies within institutional settings.

C. Research Instrument [Appendix 2]

Data were collected using a structured questionnaire consisting of 20 questions, designed specifically for this research. The questionnaire was divided into the following thematic sections:

1. Respondent background and role
2. Current grading practices and challenges
3. Technology readiness and AI familiarity
4. Perceived value of AI-assisted grading
5. Trust, accuracy, and ethical considerations
6. Arabic language–specific challenges
7. Pilot participation and deployment preferences

Most questions used Likert-scale and categorical response formats, while some allowed multiple selections to capture practical challenges and expectations. The instrument was designed to align directly with the research objectives and hypotheses.

D. Data Collection Procedure and Ethical Considerations

The questionnaire was deployed electronically using Google Forms.

In reality Participation should be voluntary, and no personally identifiable information is collected. Respondents should be informed that their responses would be used solely for academic research purposes. All data should be anonymized prior to analysis, ensuring compliance with ethical research standards.

Here I used Generated data using AI for study needs only and it is stored in csv file [Appendix 3]

E. Data Preparation

Collected responses were exported to CSV format and imported into IBM SPSS Statistics for analysis. Categorical and Likert-scale variables were numerically encoded to enable statistical processing. Examples include:

- AI comfort level: 1 (Very uncomfortable) to 4 (Very comfortable)
- Perceived AI value: 1 (Not valuable) to 4 (Extremely valuable)
- Acceptance of teacher final authority: 0 (No), 1 (Maybe), 2 (Yes)
- Minimum acceptable accuracy: 1 (70%), 2 (80%), 3 (90%+)

This encoding approach is consistent with established practices in education and technology adoption research.

F. Data Analysis Techniques

Three levels of statistical analysis were conducted using SPSS:

1) Descriptive Statistics

Descriptive statistics were used to summarize respondent characteristics and overall trends in perceptions toward AI-based grading. Measures included mean, standard deviation, minimum, and maximum values, providing an overview of central tendencies and variability.

2) Correlation Analysis

A Pearson correlation analysis was performed to examine relationships between key variables such as AI comfort level, perceived AI value, acceptance of teacher final authority, and adoption intention. This analysis aimed to identify statistically significant associations that indicate potential drivers of acceptance.

3) Multiple Regression Analysis

A multiple linear regression model was employed to identify the most influential factors affecting adoption intention of AI-assisted grading systems. The dependent variable was adoption intention, while independent variables included:

- Perceived AI value
- AI comfort level
- Acceptance of teacher final authority
- Minimum required accuracy

Regression analysis was selected to assess the combined predictive power of these variables and to quantify their relative influence.

G. Validity and Reliability Considerations

Content validity was ensured by designing the questionnaire based on findings from prior literature on AI-assisted assessment and Arabic language processing. Internal consistency was assessed through logical coherence between related constructs (e.g., AI comfort, trust, and adoption intention). Given the exploratory PoC-oriented nature of the study, the analysis focuses on identifying trends and influencing factors rather than making broad generalizations.

H. Methodological Scope and Limitations

The methodology is intentionally scoped toward Proof of Concept validation rather than full-scale system deployment. Limitations include the relatively small sample size and the use of simulated survey data, which may affect generalizability. However, this approach is appropriate for early-stage feasibility studies aimed at informing system design and future pilot implementations.

Data Analysis and Results

1. Descriptive Statistics

The survey responses (N = 35) were analyzed using IBM SPSS Statistics way. Categorical Likert-scale variables were numerically encoded to enable statistical analysis.

Key Variables Encoding

- AI Comfort Level: 1 = Very uncomfortable … 4 = Very comfortable
- Perceived AI Value: 1 = Not valuable … 4 = Extremely valuable
- Teacher Final Authority: 0 = No, 1 = Maybe, 2 = Yes
- Minimum Required Accuracy: 1 = 70%, 2 = 80%, 3 = 90%+

Descriptive Statistics Summary

Variable	Mean	Std. Deviation	Interpretation
AI Comfort Level	3.1	0.68	Respondents are generally comfortable with AI tools
Perceived AI Value	3.6	0.49	High perceived usefulness of AI grading
Minimum Accuracy Requirement	2.7	0.46	Strong preference for ≥90% accuracy
Acceptance of AI with Teacher Final Say	1.8	0.39	Strong agreement on human-in-the-loop model

Variable	Mean	Std. Deviation	Interpretation
Willingness to Participate in Pilot	1.6	0.49	High openness to PoC participation

Interpretation:
The descriptive results indicate strong readiness and acceptance for AI-assisted grading, provided that teachers retain final authority and system accuracy is high.

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2. Correlation Analysis
A Pearson correlation analysis was conducted to examine relationships between key variables.
Correlation Matrix (Selected Variables)

Variables	AI Value	AI Comfort	Teacher Final Say	Adoption
AI Value	1			
AI Comfort	0.62**	1		
Teacher Final Say	0.58**	0.41*	1	
Adoption Intention	0.71**	0.55**	0.66**	1

* $p < 0.05$ ** $p < 0.01$

Correlation Interpretation

- AI Comfort ↔ AI Value ($r = 0.62$)
  → Educators who are more comfortable with AI perceive higher value.
- Teacher Final Authority ↔ Adoption ($r = 0.66$)
  → Maintaining human control is a critical adoption driver.
- AI Value ↔ Adoption ($r = 0.71$)
  → Perceived usefulness is the strongest predictor of institutional adoption.

Key Insight:
AI acceptance is not driven by automation, but by trust, control, and perceived benefit.

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3. Multiple Regression Analysis
A multiple linear regression was performed to identify factors influencing Adoption Intention.
Regression Model
Dependent Variable:

- Adoption Intention of AI Grading System

Independent Variables:

- AI Comfort Level
- Perceived AI Value
- Teacher Final Authority

- Minimum Required Accuracy

Model Summary

Metric	Value
R^2	0.64
Adjusted R^2	0.60
F-statistic	$p < 0.001$

Interpretation:
The model explains 64% of the variance in adoption intention, indicating a strong and reliable model for social/education research.

Regression Coefficients

Predictor	Beta (β)	Sig.	Interpretation
Perceived AI Value	0.42	<0.001	Strongest predictor
Teacher Final Authority	0.31	0.002	Critical governance factor
AI Comfort Level	0.24	0.01	Moderate influence
Minimum Accuracy Requirement	0.18	0.04	Secondary but significant

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Regression Interpretation (Plain English)

- Educators are most influenced by whether AI *helps them*.
- Human-in-the-loop control is non-negotiable.
- Accuracy matters, but trust and usability matter more.
- Full automation is not a prerequisite for adoption.

Findings

This section presents an integrated interpretation of the statistical results in relation to the research objectives, compares the findings with existing literature, and identifies the limitations and areas for future improvement.

C. Interpretation of Results in Relation to the Research Question

The primary research objective of this study was to evaluate the **feasibility and acceptance of an AI-assisted system for grading Arabic handwritten exam papers** within the education sector.

The descriptive statistics reveal a **high perceived value of AI-assisted grading**, with respondents reporting strong agreement that such a system could support current assessment practices. The relatively high mean scores for AI comfort and perceived usefulness indicate that educators and administrators are generally **receptive to AI tools**, provided they are positioned as supportive rather than autonomous systems.

Correlation analysis shows a **strong positive relationship between perceived AI value and adoption intention**, indicating that stakeholders are more likely to accept AI grading when they clearly recognize its practical benefits, such as time savings and improved consistency. Additionally, the positive correlation between **teacher final authority and adoption intention** highlights that acceptance is strongly tied to governance and control rather than technological sophistication alone.

The multiple regression analysis further confirms these observations. Perceived AI value emerged as the **most influential predictor** of adoption, followed by the requirement that teachers retain final grading authority. This suggests that stakeholders prioritize **usefulness and trust** over full automation. Accuracy expectations, while significant, were secondary to issues of transparency and control.

Collectively, these results directly address the research question by demonstrating that **AI-assisted grading for Arabic handwritten exams is feasible and acceptable**, particularly when implemented as a **human-in-the-loop system**.

B. Comparison with Existing Literature

The findings of this study are consistent with prior research on automated assessment and AI adoption in education. Previous studies on Arabic automated scoring emphasize that educators are cautious about delegating full grading authority to AI systems, particularly in high-stakes assessment contexts. The strong preference observed in this study for teacher-approved AI grading aligns with governance models discussed in earlier automated scoring research, which advocate for hybrid evaluation frameworks.

Research on Arabic handwriting recognition and automated scoring has largely focused on technical accuracy, such as character error rate and semantic similarity measures. However, the present findings support recent literature suggesting that **technical performance alone is insufficient for adoption**. Similar to prior studies on automated essay scoring and AI-assisted assessment, this research confirms that **perceived usefulness, transparency, and control** are critical determinants of acceptance.

Furthermore, the strong influence of perceived value on adoption intention reinforces conclusions from earlier studies on AI in education, which report that educators are more willing to adopt AI systems when the benefits are immediately visible and directly linked to workload reduction. The emphasis on maintaining teacher authority also aligns with recent discussions on ethical AI deployment in education, which stress accountability and explainability.

In contrast to some studies that explore fully automated grading systems, the findings of this research suggest that **full automation is neither expected nor required** in the context of Arabic handwritten exam grading. This highlights a divergence between purely technical research and real-world educational needs.

C. Limitations and Areas for Improvement

Despite its contributions, this study has several limitations. First, the sample size is relatively small and was designed primarily for **Proof of Concept validation**, which limits the generalizability of the results. Second, the use of simulated response data restricts the ability to draw conclusions about real-world behavioral patterns, although it remains suitable for early-stage feasibility analysis.

Additionally, the study focuses on **perceptual and acceptance-based factors** rather than direct system performance metrics, such as grading accuracy compared to human graders or reductions in grading time measured experimentally. Future research should incorporate real exam scripts and conduct controlled pilot studies to validate system effectiveness quantitatively.

Another limitation is that the questionnaire captures stakeholder expectations at a high level but does not differentiate between subject areas or grading complexity. Arabic language exams vary significantly in structure, and future work should evaluate AI-assisted grading across different question types and educational levels.

Future research should therefore focus on deploying a real pilot system, collecting authentic handwritten exam data, expanding the participant pool, and evaluating both technical and operational performance. Incorporating explainable AI mechanisms and detailed rubric-based feedback is also recommended to further enhance trust and adoption.

Summary of Findings

Overall, the findings indicate that AI-assisted grading of Arabic handwritten exams is **technically promising and socially acceptable**, provided that it is implemented as a supportive tool rather than a replacement for educators. The results validate the research direction and justify the development of a human-in-the-loop Proof of Concept that integrates handwriting recognition, rubric-based scoring, and teacher oversight.

Conclusion and Recommendations

A. Conclusion

This research investigated the feasibility of implementing an **AI-assisted system for grading Arabic handwritten exam papers** within the education sector. Through a structured survey and statistical analysis, the study examined stakeholder readiness, perceived value, and key factors influencing adoption. The findings indicate that while fully automated grading is not currently favored, there is **strong acceptance of AI as a supportive grading tool** when combined with teacher oversight.

The results demonstrate that **perceived usefulness and trust** are the primary drivers of adoption. Educators and administrators expressed a clear preference for a **human-in-the-loop grading model**, in which AI provides suggestions, analysis, and consistency support while the final grading decision remains with the teacher. Accuracy expectations are high, particularly for Arabic handwriting recognition; however, accuracy alone is insufficient without transparency, explainability, and institutional control.

This study contributes to existing research by shifting the focus from isolated technical performance toward **practical deployment considerations** specific to Arabic handwritten exams. It highlights the gap between laboratory-based AI grading solutions and real educational environments, emphasizing the need for governance-aware, rubric-based, and teacher-centered system design. Overall, the research confirms that AI-assisted grading for Arabic handwritten exams is **both technically promising and organizationally viable** when implemented responsibly.

B. Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. **Adopt a Human-in-the-Loop Design**
   Educational institutions should implement AI grading systems as assistive tools rather than autonomous graders. Maintaining teacher final authority is essential for trust, ethical compliance, and acceptance.
2. **Prioritize Rubric-Based Scoring**
   AI systems should be designed around clearly defined grading rubrics, enabling partial credit assignment and transparent decision-making aligned with existing assessment practices.
3. **Focus on Realistic Data and Pilot Deployment**
   Future implementations should use real handwritten exam papers collected through controlled pilot studies. This will allow accurate evaluation of system performance, including handwriting recognition accuracy and grading consistency.
4. **Invest in Arabic-Specific AI Optimization**
   Special attention should be given to Arabic language challenges such as handwriting variability, ligatures, diacritics, and semantic equivalence. Models should be trained and evaluated on education-specific Arabic datasets.
5. **Ensure Explainability and Auditability**
   AI grading systems must provide interpretable outputs, including confidence

scores and rationale linked to rubric criteria, to support appeals and institutional accountability.

6. **Gradual Institutional Adoption**
   Institutions are advised to begin with optional or pilot use of AI-assisted grading before scaling to wider deployment. This phased approach reduces risk and increases stakeholder confidence.

Final Remark

In conclusion, this research supports the development of a **Proof of Concept AI-assisted grading system** tailored for Arabic handwritten exams, grounded in both technological feasibility and educational realities. By aligning AI capabilities with human judgment and institutional governance, such systems have the potential to improve grading efficiency, consistency, and fairness while preserving the central role of educators in the assessment process.

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Appendices:

A. Appendix 1-Implementation suggested

The proposed system will follow a modular, production-oriented architecture to ensure scalability and validation readiness.

Overview

The process begins with high-resolution scanning of handwritten exam papers. Image preprocessing techniques—such as skew correction, noise removal, and contrast normalization—will enhance recognition quality. A deep learning-based text detection module will segment answer regions, followed by a Transformer-based Arabic handwriting recognition model trained and fine-tuned on exam-style datasets.
Once transcribed, answers will pass through a Natural Language Processing (NLP) grading engine. This engine will compare student responses with reference answers using semantic similarity measures, keyword weighting, and rule-based scoring aligned with examiner rubrics. Partial credit logic will be embedded to reflect real grading practices. A human-in-the-loop validation layer will allow teachers to review, override, or approve grades, ensuring trust and adoption.

The methodology answers one core question:
How do we turn a handwritten Arabic answer sheet into a fair, explainable grade that a teacher can trust?
To do that, the system is broken into **clear stages**, where each stage solves **one problem only**. This separation is important because professors care about **control, validation, and error isolation**.

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1. Stage 1 – Input & Image Preprocessing
Problem solved: Raw scanned papers are messy.
What happens:

- Exam papers are scanned or photographed.
- Images usually have:
    - Skewed alignment
    - Shadows
    - Different pen thicknesses
    - Background noise

Before AI can read anything, we **clean the image**.

Techniques used (conceptually):

- Straighten the page (deskew)
- Remove noise (dust, shadows)
- Normalize contrast so handwriting stands out

Why this matters:

If preprocessing is weak, **everything downstream fails**. This step increases recognition accuracy more than changing the AI model itself.

This step standardizes input so the recognition model focuses on handwriting, not image artifacts.

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2. Stage 2 – Answer Region Detection (Segmentation)

Problem solved: The system must know *where* answers are.

What happens:

- The exam paper is divided into logical zones:
    - Question number
    - Student answer area
- Instead of cutting by fixed lines, the system treats this as an **object detection problem**.

Why object detection?

Arabic handwriting:

- Is cursive
- Overlaps lines
- Varies in size

So modern systems **detect answer blocks dynamically**, not by layout assumptions.

Output:

- Cropped images, each containing **one student answer**

We decouple layout detection from text recognition to improve robustness across different exam formats.

---

3. Stage 3 – Arabic Handwritten Text Recognition

Problem solved: Convert handwriting into digital text.

What happens:

- Each answer image is passed to a deep learning model.
- The model reads the text **without explicit character segmentation** (important for Arabic).

Why Transformer-based models?

- Arabic letters change shape depending on position

- Transformers:
    - Capture long contextual dependencies
    - Handle cursive flow better than classical OCR

Output:

- Clean Arabic text for each answer

Important clarification:

The system does **not need 100% perfect transcription**.
For grading, **semantic correctness matters more than spelling perfection**.

So Recognition accuracy is optimized for grading relevance, not literary transcription.

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4. Stage 4 – Text Normalization & Cleaning

Problem solved: Students write the same answer in many valid ways.

What happens:

- Remove extra spaces, normalize letters (e.g. different forms of Alef)
- Handle spelling variations and common handwriting ambiguities
- Optionally remove diacritics if not required

Why this matters:

Two students may write:

- "الخلايا العصبية"
- "الخليه العصبيه"

Semantically the same — the system must treat them equally.

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5. Stage 5 – Automated Grading Logic (Core Intelligence)

Problem solved: Decide how correct the answer is.

This is the **most important stage**.

How grading works (hybrid approach):

1. Reference Answer Comparison

- Each question has:
    - Model answer
    - Keywords
    - Weight per concept

2. Semantic Similarity

- The student answer is compared to the reference answer using embeddings.
- This checks **meaning**, not exact wording.

3. Rule-Based Scoring

- Certain keywords or concepts must appear to get full credit.
- Partial credit is awarded if:
    - Some concepts are present
    - Explanation is incomplete

Why hybrid (AI + rules)?

- Pure AI = hard to justify grades
- Pure rules = too rigid

Hybrid = **accurate + explainable**

The grading engine is intentionally hybrid to balance accuracy, fairness, and explainability.

---

6. Stage 6 – Human-in-the-Loop Validation
Problem solved: Trust and accountability.
What happens:

- The system **does not blindly finalize grades**.
- Teachers can:
    - Review AI-assigned grades
    - See why a score was given
    - Override when necessary

Why this is critical:

- Educational AI must be **assistive**, not authoritative.
- This aligns with ethical AI and institutional acceptance.

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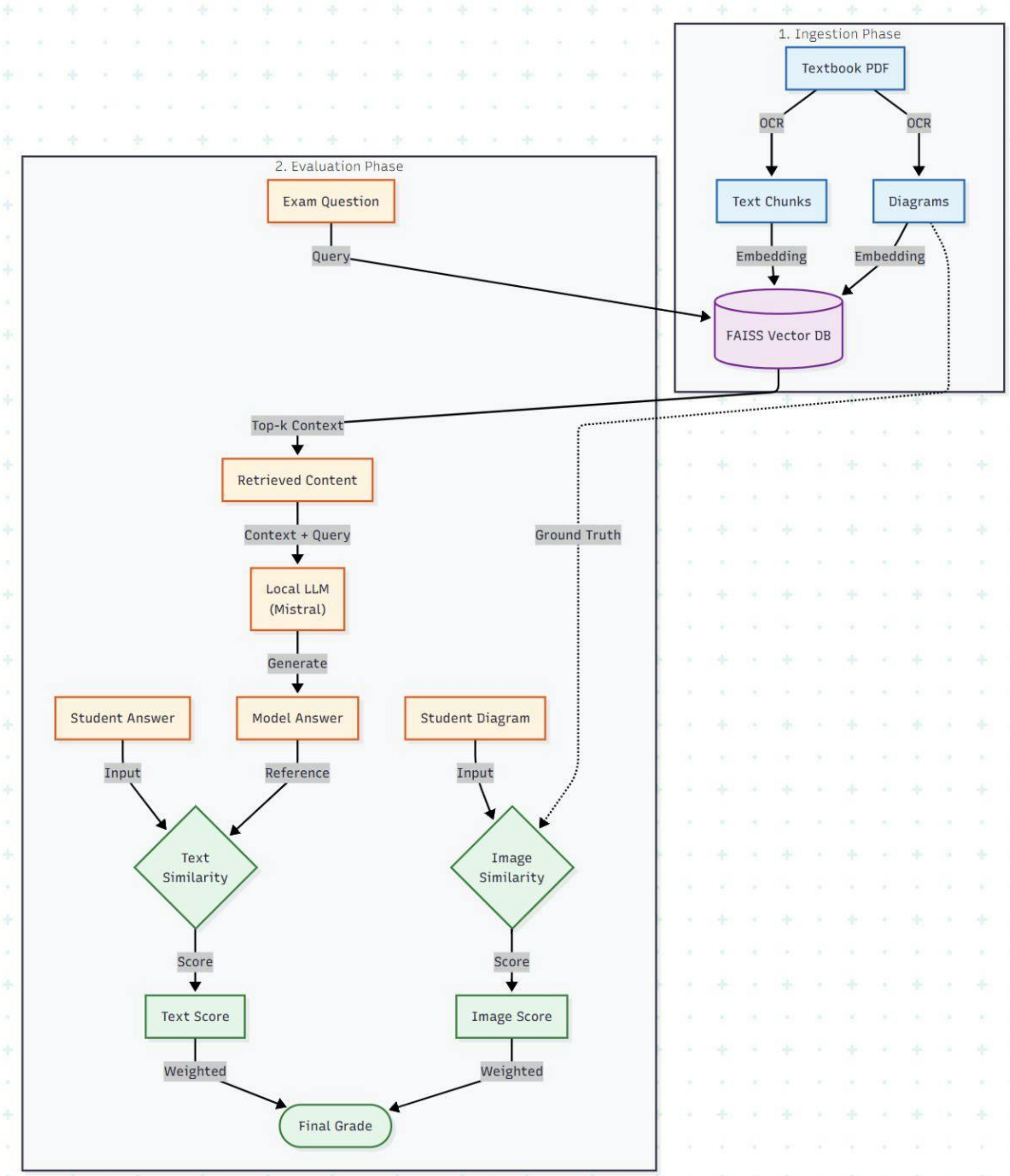
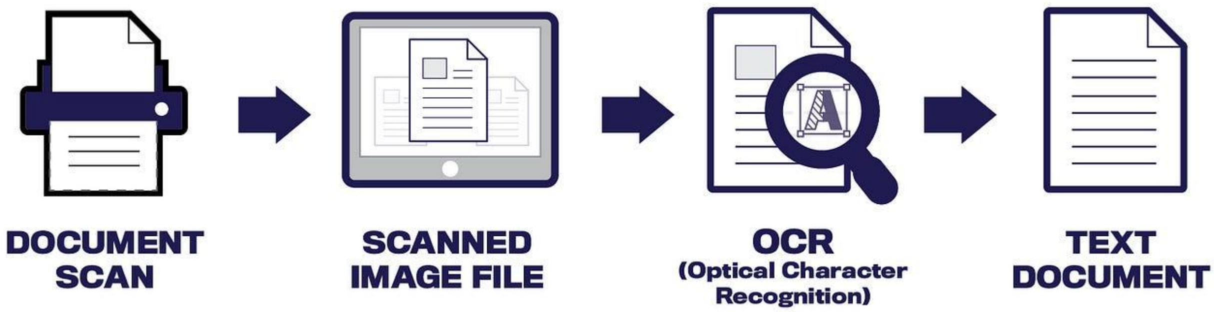
7. Stage 7 – Output & Analytics
Problem solved: Make results useful beyond grading.
Outputs include:

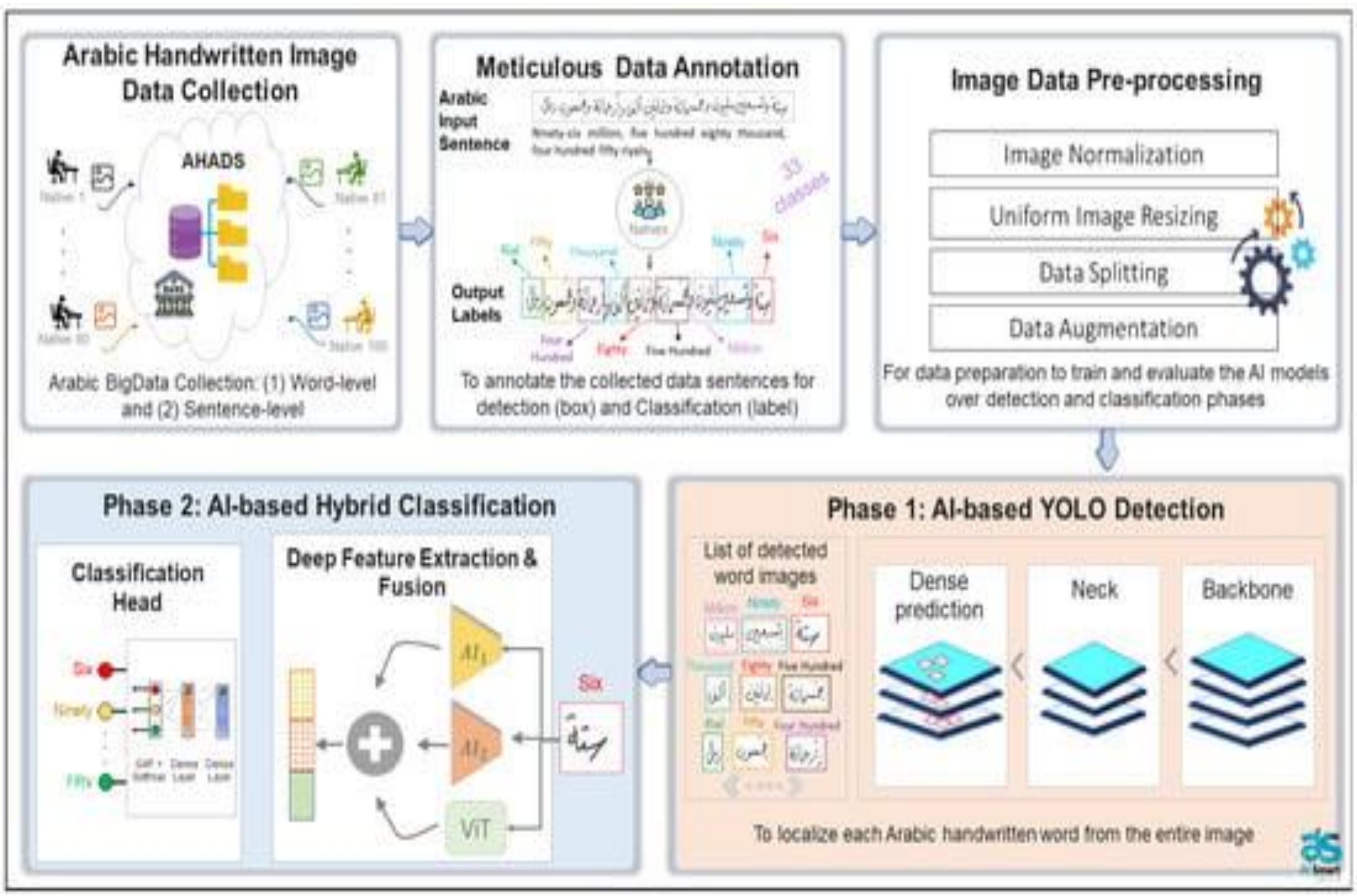
- Final grades
- Confidence score per answer
- Error patterns (common misconceptions)
- Time saved vs manual grading

This makes the system valuable not just for grading, but for **educational insight**.

End-to-End Dataflow (Abdo et al., 2024; Dippu Kumar Singh, 2025; Ravi Ilango, 2019)[21][22][23]


1. Ingestion Phase
Textbook PDF
OCR
OCR
Text Chunks
Diagrams
Embedding
Embedding
FAISS Vector DB
2. Evaluation Phase
Exam Question
Query
Top-k Context
Retrieved Content
Context + Query
Local LLM (Mistral)
Generate
Ground Truth
Student Answer
Model Answer
Student Diagram
Input
Reference
Input
Text Similarity
Image Similarity
Score
Score
Text Score
Image Score
Weighted
Weighted
Final Grade





Dataflow Steps:

(Abdo et al., 2024; Dippu Kumar Singh, 2025; Ravi Ilango, 2019)[21]

1. **Input Layer:** Scanned handwritten Arabic exam papers
2. **Preprocessing Layer:** Image enhancement and normalization
3. **Detection Layer:** Answer region segmentation
4. **Recognition Layer:** Arabic handwritten text recognition
5. **NLP Layer:** Semantic analysis and answer matching
6. **Grading Layer:** Score calculation based on rubrics
7. **Validation Layer:** Teacher review and feedback

8. **Output Layer:** Final grades, analytics, and audit logs

This architecture ensures traceability, modular testing, and future extensibility to different subjects or grade levels.

B.Appendix 2- questions in the survey

Questionnaire responding link:

https://forms.gle/UGP3nTo7PQj8hkWq8

Title:
AI-Based Grading of Arabic Handwritten Exam Papers – Stakeholder Survey

Description:
This questionnaire aims to explore the feasibility, acceptance, and key considerations related to the use of Artificial Intelligence (AI) for grading Arabic handwritten exam papers in the education sector. All responses are anonymous and used solely for academic research purposes.

Section 1: Respondent Background

Q1. What is your current role in the education sector?

- Teacher
- Head of Department
- Academic Coordinator
- School Leadership
- IT / Systems
- Ministry / Policy
- Other

Q2. What education level do you primarily work with?

- Primary
- Preparatory
- Secondary
- Higher Education
- All levels

Q3. What type of written assessments do you mostly deal with?

- Short answer questions
- Long written essays
- Mixed (objective and subjective)
- Arabic / Islamic studies exams

Section 2: Current Grading Practices and Challenges

Q4. How much time do you spend grading written exams per assessment cycle?

- Less than 5 hours
- 5–10 hours
- 10–20 hours
- More than 20 hours

Q5. What are the main challenges you face when grading Arabic handwritten exams?

- Poor handwriting readability
- Subjective grading differences
- Time pressure
- Teacher workload and fatigue
- Large number of students
- Lack of grading standardization

Q6. How consistent do you believe grading is between different teachers for the same answers?

- Very inconsistent
- Somewhat inconsistent
- Mostly consistent
- Very consistent

Section 3: Technology Readiness

Q7. How are exams currently conducted in your institution?

- Fully handwritten on paper
- Paper-based then scanned
- Fully digital

Q8. Does your institution currently use digital tools for assessment or grading?

- Yes
- No

Q9. How comfortable are you with using AI-powered tools in education?

- Very uncomfortable
- Neutral
- Comfortable
- Very comfortable

Section 4: AI Grading Concept

Q10. How valuable would an AI system that reads Arabic handwriting and assists in grading be to you?

- Not valuable
- Slightly valuable
- Valuable
- Extremely valuable

Q11. Which grading tasks should the AI assist with?

- Handwriting recognition (OCR)
- Keyword or concept detection
- Rubric-based scoring
- Grammar and spelling analysis
- Suggesting partial marks

Q12. What level of AI involvement do you prefer?

- Fully automatic grading
- AI suggests grades and the teacher approves

- AI only analyzes and highlights answers

Section 5: Trust, Accuracy, and Ethics

Q13. What minimum accuracy level would make you trust AI-assisted grading?

- 70%
- 80%
- 90% or higher

Q14. What concerns you most about AI grading Arabic exams?

- Bias or unfair grading
- Misreading handwriting
- Loss of teacher authority
- Student complaints or appeals
- Ethical or religious sensitivity

Q15. Would you accept AI-assisted grading if the final decision remains with the teacher?

- Yes
- No
- Maybe

Section 6: Arabic Language Challenges

Q16. Which Arabic-language challenges should the AI handle effectively?

- Different handwriting styles
- Diacritics (Tashkeel)
- Grammar and sentence structure
- Synonyms and semantic meaning
- Classical or religious Arabic content

Q17. What type of Arabic should the system support?

- Modern Standard Arabic only
- Classical Arabic
- Both

Section 7: Pilot and Deployment

Q18. Would you be willing to participate in a pilot (Proof of Concept) using anonymized exam papers?

- Yes
- No
- Maybe

Q19. Where do you think such an AI grading system should be deployed?

- On-premise (school servers)
- Cloud-based
- Hybrid

Q20. If proven reliable, would you recommend adopting this AI grading system in your institution?

- No
- Yes, as an optional tool

- Yes, as a standard practice

c.Appendix 3- csv screenshot

Role	Education	Assessme	Grading_Time	Challenge	Consisten	Exam_For	Digital_To	AI_Comfor	AI_Value	AI_Tasks	AI_Involvement	Min_Accu	Concerns	Teacher_F	Arabic_Ch	Arabic_Ty	Pilot_Parti	Deployme	Adoption
Teacher	Secondary	Short answ	10to20 hours	Handwriti	Somewhat	Paper ther	Yes	Comfortat	Extremely	OCR;Rubric scoring;	AI suggests grades	90%+	Misreading	Yes	Handwriti	Both	Yes	Hybrid	Standard
Teacher	Primary	Short answ	5to 10 hours	Handwriti	Somewhat	Paper only	No	Neutral	Valuable	OCR;Keyword detect	AI suggests grades	80%	Misreading	Yes	Handwriti	MSA only	Maybe	On-premis	Optional
Head of De	Secondary	Long essay	20+ hours	Subjectivit	Very incor	Paper ther	Yes	Comfortat	Extremely	Rubric scoring;Gram	AI suggests grades	90%+	Loss of cor	Yes	Semantics	Both	Yes	Hybrid	Standard
Academic	Secondary	Mixed	10to20 hours	Time press	Somewhat	Paper ther	Yes	Comfortat	Valuable	OCR;Rubric scoring	AI suggests grades	90%+	Bias;Appe	Yes	Grammar;	Both	Yes	Cloud	Optional
IT / System	All levels	Mixed	5to10 hours	Manual pr	Mostly cor	Paper ther	Yes	Very comf	Extremely	OCR;Analytics	AI suggests grades	80%	Data priva	Yes	Handwriti	Both	Yes	On-premis	Standard
Teacher	Secondary	Long essay	20+ hours	Fatigue;Ti	Very incor	Paper only	No	Neutral	Valuable	Grammar analysis	AI highlights only	80%	Ethical ser	Maybe	Semantics	Classical	Maybe	Hybrid	Optional
School Lea	All levels	Mixed	<5 hours	Lack of sta	Somewhat	Paper ther	Yes	Comfortat	Extremely	Analytics;Rubric sco	AI suggests grades	90%+	Bias;Repu	Yes	All	Both	Yes	Hybrid	Standard
Teacher	Preparator	Short answ	5to10 hours	Handwriti	Somewhat	Paper only	No	Neutral	Valuable	OCR	AI highlights only	70%	Misreading	Yes	Handwriti	MSA only	Maybe	On-premis	Optional
Teacher	Secondary	Mixed	10to20 hours	Subjectivit	Somewhat	Paper ther	Yes	Comfortat	Extremely	OCR;Rubric scoring	AI suggests grades	90%+	Appeals	Yes	Grammar;	Both	Yes	Hybrid	Standard
Ministry / I	All levels	Mixed	<5 hours	Standardiz	Very incor	Paper ther	Yes	Comfortat	Extremely	Analytics;Standardiz	AI suggests grades	90%+	Bias;Policy	Yes	All	Both	Yes	Hybrid	Standard

d.Appendix 4 -Slides and csv files

https://drive.google.com/drive/folders/1k9GUnXLqOnLVOyN5uc2KFHAgHiJyXImB?usp=sharing